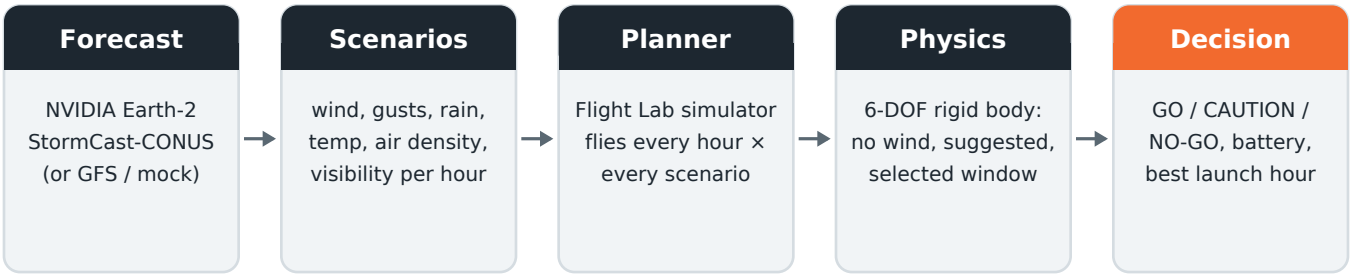
Drone Scout

Flight Lab + Mission Planner · pre-flight decisions for crop-survey drones

What the project does

A grower or researcher wants to survey a field with a drone in two or three days. Before they drive out, Drone Scout answers: **will the drone finish the survey safely at that time, how much battery will it take, and is there a better hour?** It forecasts the weather at the field, flies the real survey route in a simulator in every forecast scenario, and turns the result into a GO / CAUTION / NO-GO decision with reasons.



What is in it

- **Flight Lab**: a quadcopter simulator (DJI Phantom 4 Multispectral reference) flying lawn-mower survey lanes over real UF/IFAS-supplied field imagery, with wind, gusts, GNSS noise, battery and camera triggers.
- **Weather**: NVIDIA Earth-2 **StormCast-CONUS** (AI, 3 km, hourly, added to Earth2Studio in July 2026) steered by NOAA GFS for days-ahead plans; GFS alone without a GPU; a mock forecast for demos.
- **Mission planner**: flies every launch hour × every forecast scenario, finds the best hour and the battery needed.
- **6-DOF physics**: a rigid-body model (attitude, four motors, flight controller, Dryden turbulence, RK4 at 500 Hz) that shows how wind pushes the drone off its path and how it tilts and rotates, checked against hand calculations.
- **Dust and visibility**: blowing dust and rain lower visibility; under the FAA 3-statute-mile minimum (14 CFR 107.51(c)) it is NO-GO.

How to read the map

- **Green**: original path with no wind
- **Orange**: suggested GO path for the day (best launch hour, typical scenario)
- **Blue**: the window you selected, and the live simulator flight

Every planner and physics view shows these three paths, so you see at a glance how your window compares with calm air and with the best hour.

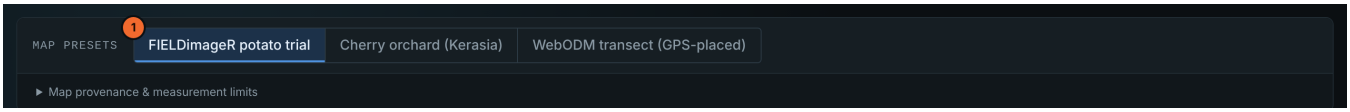
Run it

```
cd dronesim && python -m http.server 8000 # then open http://127.0.0.1:8000/index.html
# or double-click index.html and use Load forecast (button 8) · tests: node
tests/physics.test.cjs
```

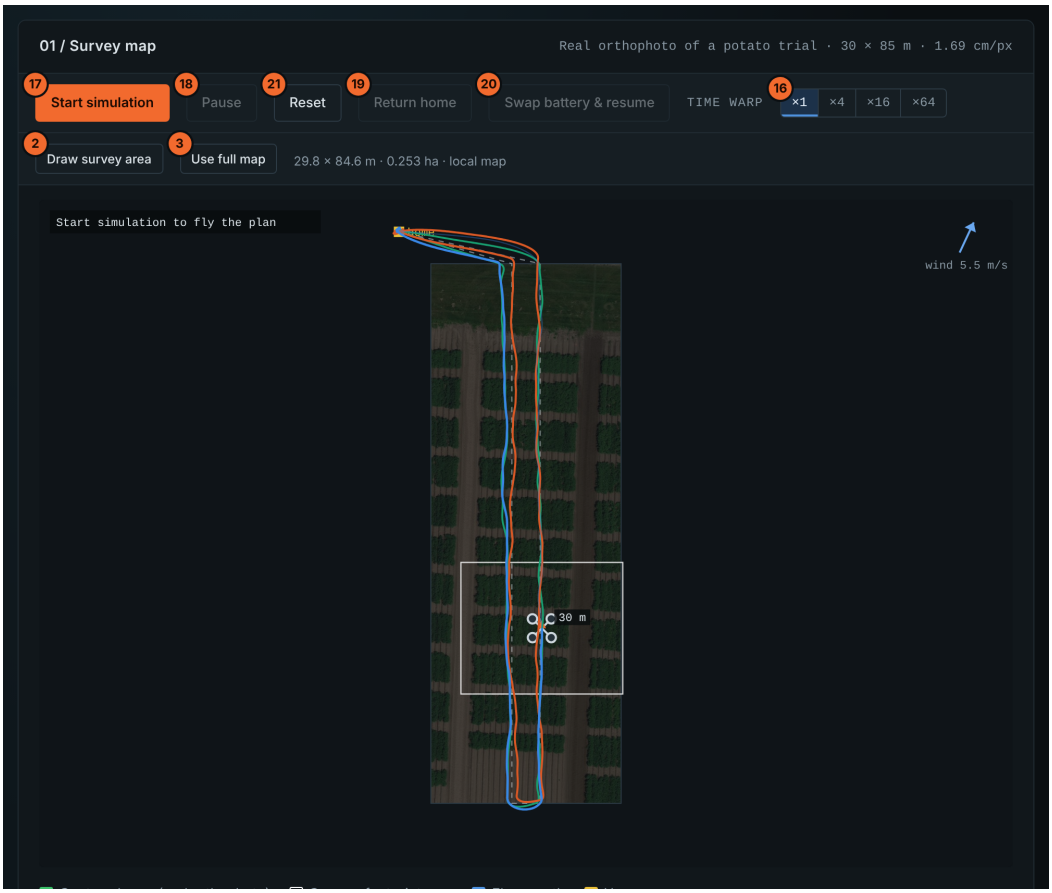
Every button, in the order you use it

The orange numbers on the screenshots match the steps. A normal run is steps 1 → 3, then 8 → 15, then 22 → 27. Steps 4–7 are only needed to change the aircraft; 16–21 control a live flight; 29–32 export a live flight.

Stage A · Choose the field

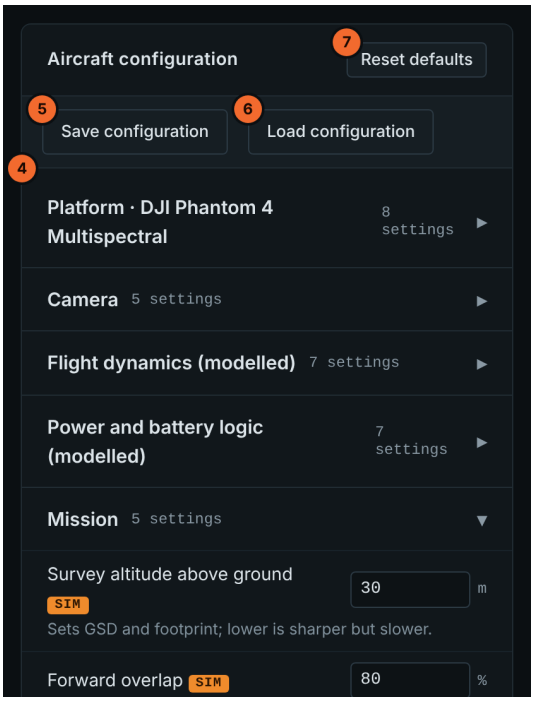


#	Button	What it does	When
1	**Map presets**	Pick the field: potato trial, cherry orchard or WebODM transect.	First, every session



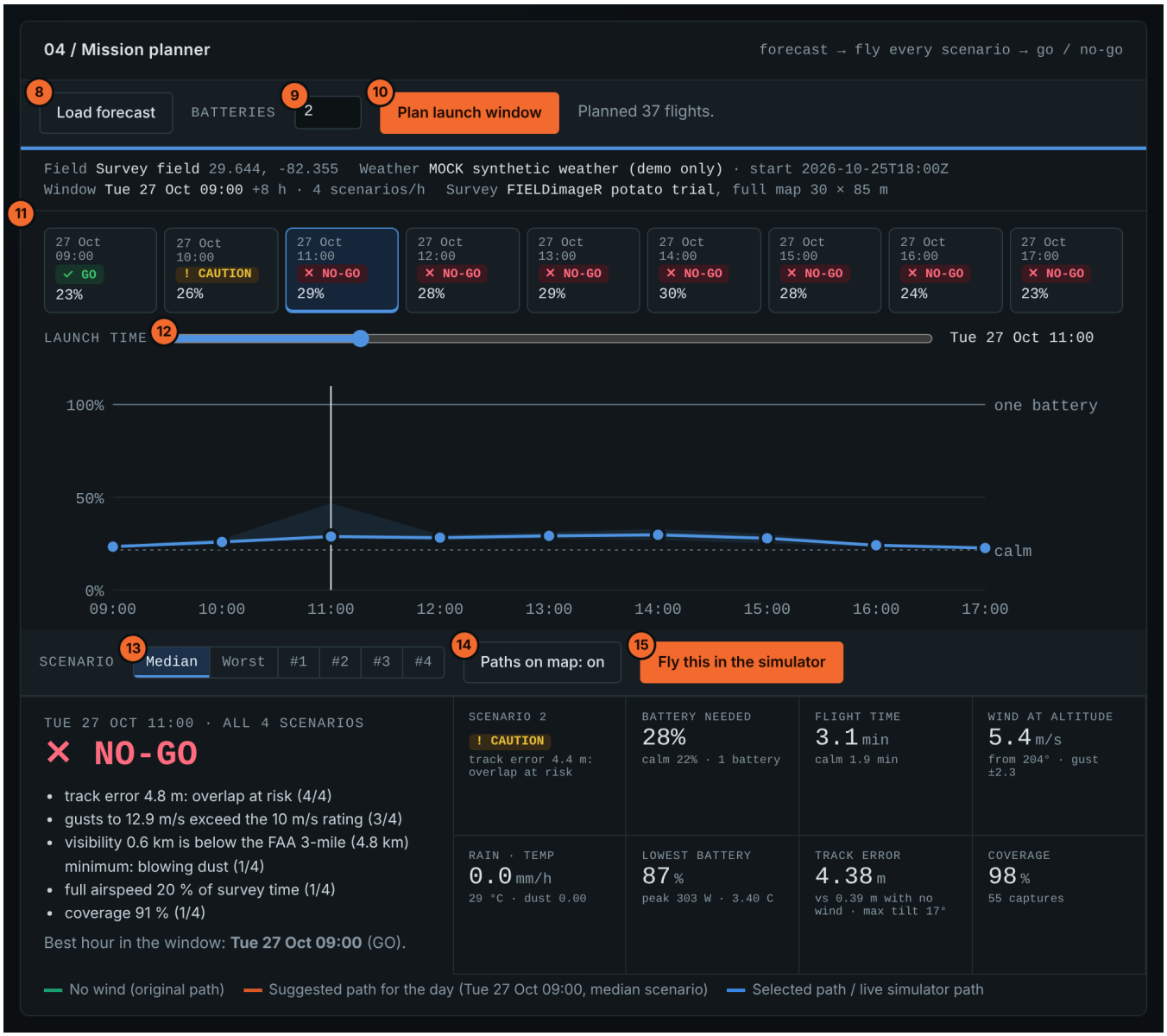
#	Button	What it does	When
2	**Draw survey area**	Drag a rectangle on the map to survey only part of the field.	When you want a portion
3	**Use full map**	Go back to surveying the whole field.	To undo step 2

Stage B · Aircraft (optional)



#	Button	What it does	When
4	**Aircraft settings**	Altitude, overlap, speeds, battery, reserve, wind sensors. Tags show SPEC / DERIVED / ASSUMED / SIM.	Only to change the drone or survey
5	**Save configuration**	Download the current settings as JSON.	Before experimenting
6	**Load configuration**	Load a saved JSON (also the replay files from the planner).	To restore or replay
7	**Reset defaults**	Back to the P4 Multispectral reference values.	If settings get messy

Stage C · Plan the launch window (04 / Mission planner)



#	Button	What it does	When
8	**Load forecast**	Pick forecast_scenarios.json (loads by itself when the page is served).	Only if opened as a file
9	**Batteries**	How many batteries you will carry.	Before planning
10	**Plan launch window**	Flies the survey in every hour × scenario with your settings and drawn area.	Main step
11	**Hour tiles**	Each launch hour with its decision and battery needed. Click one to select it.	To compare hours
12	**Launch time slider**	Same as the tiles; also ← / → keys. The map redraws the blue path.	To scan the day
13	**Scenario**	Median, Worst, or one forecast scenario (#1…#n) for that hour.	To see the spread

#	Button	What it does	When
14	**Paths on map**	Show or hide the green / orange / blue paths on the survey map.	For a clean map
15	**Fly this in the simulator**	Puts that hour’s weather into the aircraft and flies it live in Flight Lab.	To watch the flight

Stage D · Watch a live flight (01 / Survey map)

#	Button	What it does	When
16	**Time warp ×1 ×4 ×16 ×64**	Playback speed of the live flight.	Use ×16–×64 for demos
17	**Start simulation**	Fly the current plan with the current settings.	After 15, or on its own
18	**Pause / Resume**	Freeze and continue the live flight.	To explain a moment
19	**Return home**	Abort the survey and fly home.	To show a manual abort
20	**Swap battery & resume**	After a battery return, continue with a fresh pack.	Big fields
21	**Reset**	Clear the live flight.	Before a new run

Stage E · Physics (05 / Physics 6-DOF)



#	Button	What it does	When
22	**Run 6-DOF physics**	Flies the survey as a rigid body three times: no wind, suggested GO path, selected window.	After choosing an hour
23	**Play / Pause**	Plays the physics flight on the map and the 3D aircraft view (roll, pitch, heading).	To show motion
24	**Speed ×1 ×10 ×60**	Playback speed.	
25	**Scrubber**	Jump to any moment; the charts' cursor follows. You can also click a chart.	To find gust hits
26	**Show on map**	Show or hide the physics paths and drone on the map.	

#	Button	What it does	When
27	**Download CSV (MATLAB)**	All three flights, 20 samples/s: position, roll/pitch/yaw, rates, motor thrust, power, wind, off-path distance.	For MATLAB: plot_physics(file)
28	**Model, equations…**	Equations, controller, airframe values with their basis.	For technical judges

Stage F · Export a live flight (03 / Survey captures)



#	Button	What it does	When
29	**Export mission**	Full mission JSON: telemetry, captures, configuration, model notes.	After a live flight
30	**Download CSV**	One row per camera capture.	
31	**Footprints JSON**	Capture footprints in local map metres.	
32	**Show text to copy**	The same data as text, if downloads are blocked.	

How the decision is made

Each forecast scenario gets a decision; an hour is NO-GO if half or more of its scenarios are NO-GO, CAUTION if any is NO-GO or half are CAUTION, otherwise GO. The suggested GO path is the median scenario of the best hour.

Check	NO-GO when	CAUTION when	Source
Wind and gusts	mean + 2 × gust σ above 10 m/s	over 5 % of survey time at full airspeed; tilt over 22°	DJI P4M spec; simulator
Rain	over 0.5 mm/h	some scenarios show showers	radar reflectivity → rain rate
Visibility (dust, rain)	under 4.8 km (3 statute miles)	under 8 km: keep line of sight, hazy images	FAA 14 CFR 107.51(c); estimate
Temperature	outside 0–40 °C	under 10 °C: warm the batteries	DJI P4M spec
Battery	more batteries needed than carried; emergency landing	uses every battery; lands near the reserve	simulator
Tracking	—	track error over 2 m (photo overlap at risk); coverage under 95 %	simulator

Why dust matters

Dust barely changes how the drone flies, but it can make flying illegal and the images useless. Blowing dust and rain both scatter light, so their effects add. When estimated visibility falls under the FAA minimum of 3 statute miles from the control station, the hour is NO-GO even if the drone could physically fly it. In the sample forecast the 11:00 flight stays within 1 m of its path in the physics model, yet the hour is NO-GO because of dust visibility.

What is measured, derived and assumed

Piece	Basis
Wind, temperature, pressure, radar reflectivity	Weather model output (StormCast-CONUS or GFS)
Wind at flight altitude, gust size	Assumed profile (exponent 0.14) and turbulence intensity 0.15, plus ensemble spread
Rain rate, air density	Derived with standard formulas (Marshall–Palmer; ideal gas)
Visibility from dust and rain	Assumed empirical relation: an estimate, not a model output
Hover power vs air density, cold batteries	Derived (momentum theory) and assumed (−0.75 %/°C below 20 °C)
6-DOF airframe: inertia, drag, motor lag, thrust-to-weight	Assumed; mass, size and propellers follow the P4 Multispectral
Physics accuracy	Hover thrust = mg/4, hover power = 177 W, steady-wind tilt = atan(drag/weight) to 0.01°, power ∝ 1/√ρ

Best things to do next

Ranked by how much they help the hackathon result for the time they take.

Before the event (tonight)

#	Do this	Why
1	**Run one real forecast on HiPerGator** for a real UF field and the demo date: `python earth2_weather.py --source stormcast --lat … --lon … --mission-time "…" --members 4`. Keep the output and demo from it (no live GPU needed on stage).	Turns the demo from mock to real AI weather; judges will ask.
2	**Check the hackathon rules on existing code.** Present Flight Lab as the starting platform and make the forecast, planner and physics clearly the event work.	Avoids disqualification.
3	**Rehearse a 3-minute demo** (script below) on the computer you will present from.	Live demos fail on setup, not on code.
4	**Know the limits** (see “How the decision is made”) for Q&A: what is model output, what is estimated.	Honest answers score better than big claims.

During the event (pick two or three)

Idea	Effort	Impact
Wind-aware return home. The simulator decides when to go home using calm-air power, which is why strong headwinds end in emergency landings. Make the reserve include the headwind, then show the emergency landings disappear.	Medium	High: "our simulator found a real safety bug"
Explain the decision with Gemma (MLH Best Use of Gemma): feed the hour’s reasons and numbers, get a plain-language briefing a grower can read.	Medium	High: visible AI + sponsor prize
Hindcast check: run the forecast for a past windy day and compare with what was observed (HRRR analysis or a weather station).	Medium	High credibility
Real field outline and location for a UF/IFAS plot instead of a local map.	Low	Medium
Twinly: when the plan says NO-GO, let the desktop agent reschedule the calendar event and notify the team.	Low	Sponsor prize
After the flight: run a crop-health model on the captured images (your later phase). Keep for after the hackathon unless time is left.	High	Medium in 48 h

3-minute demo script

Time	Show	Say
0:00	Title, the survey map	"Growers plan drone surveys days ahead. Today they guess about the weather."
0:20	Draw survey area (2), Plan launch window (10)	"Earth-2 forecasts the field; our simulator flies the survey in every forecast scenario."
0:50	Drag the slider (12) through the day	"09:00 is GO. By 11:00 blowing dust drops visibility under the FAA 3-mile limit: NO-GO."
1:20	Map: green / orange / blue	"Green is calm air, orange the best path of the day, blue the window you asked for."
1:40	Run 6-DOF physics (22), Play (23)	"A full rigid-body model: watch it lean into the gusts; 41 % more energy than calm air."
2:20	Battery needed, best hour	"Fly at 09:00 with one battery. That is the decision, made two days ahead."
2:40	Tests, assumptions	"Physics checked against hand calculations; estimates are labelled."

Pitch tip: lead with the decision ("fly at 09:00, not 11:00"), then show how. One strong number beats ten features.